

Generative machine learning models for data assimilation

Our team of mentors

Alphabetically by last name:

- Ricardo Baptista (<https://www.ricardobaptista.com/>) - Postdoc at Caltech, USA
- Giulio Trigila (<https://sites.google.com/view/giuliotrigila/welcome>) - Professor at Baruch College, CUNY, USA

Broad Goal Of The Project

Machine learning aims to characterize properties of probability distributions from samples. These distributions quantify our beliefs for possible outcomes of an experiment or hypothesis such as:

- Is it likely to rain in Boston tomorrow?
- Will the Boston Red Sox win the World Series in 2022?

In practice we summarize our beliefs by measuring properties of distributions such as their moments and quantiles. This project will focus on the distributions arising from a collection of data (e.g., images, text, videos, etc.). In this project, we will build algorithms to sample these distributions using tools from optimal transport. The applications of this project range vastly from improving predictions of weather forecasting systems to financial models. For example, conditional sampling is core problem of many inference procedures that estimate the current state of atmospheric variables current limited temperature and pressure measurements. These procedures are deployed on a regular basis every day by weather prediction services. Devising robust and unbiased methods to perform these tasks will have a significant benefit for these applications.

Resources

- *Deep MMD Gradient Flow without adversarial training* by A. Galashov, V. de Bortoli, A. Gretton (2024): available at <https://arxiv.org/pdf/2405.06780>
- *Normalizing Flows for Probabilistic Modeling and Inference* by G. Papamakarios et al., (2021): available at <https://arxiv.org/pdf/1912.02762.pdf>
- *A family of nonparametric density estimation algorithms* by E.G. Tabak, C.V. Turner (2013): available at <https://core.ac.uk/download/pdf/80364696.pdf>
- *Data Driven Optimal Transport* by E.G. Tabak and G. Trigila (2016): available at <https://math.nyu.edu/~tabak/publications/Tabak-Trigila.pdf>

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This page was last edited on 5 June 2025, at 19:14.